1. What is the purpose of a loop structure?
To repeat a line of code multiple times

2. Explain the difference between a while statement and a do-while statement.
A while loop--condition is checked before the loop body runs, so if the condition is not met it will not run at all.
A do while loop--checks after the loop body is run, so it will run at least once.

3. An input validation loop is a loop that checks user input for valid data. If valid data is not entered, the loop iterates until valid data is entered. In which review of this chapter did you write code for an input validation loop?
Review: Account setup

4. a) What is an infinite loop?
a loop that never ends because its condition is never met
b) List two types of errors that can lead to an infinite loop.
The loop condition is not written properly. And the loop that has to change isn't updated properly.
c) What is meant by overflow?
You put in a value that is bigger than the maximum value.

5. How many times will the do - while loop execute? int x = 0; do { x = x + 2; while (x < 120);
It will run 60 times

6. What initial value of x would make the loop infinite?
```
do {
 x = x + 3;
 while (x < 120);
```
None, without changing the code

7. Compare and contrast counters and accumulators. List two uses for each.
A) counter will check the amount of times a loop has run.
- It will track how many people did their entry logs

B) Accumulators find the sum of a track meet run times
- Tracking total distance traveled

8)Write a for statement that sums the integers from 3 to 10,inclusive.
```
   int sum = 0;
for (int i = 3; i <= 10; i++) {
   sum += i;
}
System.out.println("Sum from 3 to 10 is: " + sum);
```

9)List two factors that should be considered when determining which loop structure to choose.
When the loop condition is checked.

Readability, for loops are easier to read
11)Consider the following assignment:
String x = "my string.";
Determine the value returned by each of the following methods:
a) x.length() -- 10
b) x.substring(0, 3) -- “my”
c) x.toLowerCase() -- “my string”
d) x.toUpperCase -- “MY STRING”
e) x.trim() -- “my string”